

Lesson 13 Normal system

Tuesday, April 28, 2026

20:57

Last time we showed if B and B' are two bases, $\exists w \in W_B$ s.t. $w(B') = B$. In fact, such a w is unique:

prop

let $w \in W_B$. Then $w(B) = B \Leftrightarrow w = \text{Id}$.

pf:

Suppose $w(B) = B$. $w =$ a product of $S_{\beta_1}, \dots, S_{\beta_n}$.

Let $w = S_{\beta_{i_1}} \dots S_{\beta_{i_m}}$ be a shortest representation.

Claim: $w(\beta_{i_m}) < 0$ (i.e. $w(\beta_{i_m}) \in \Delta_-$)

Suppose otherwise $w(\beta_{i_m}) > 0$. Since $S_{\beta_{i_m}}(\beta_{i_m}) = -\beta_{i_m} < 0$, we have that $S_{\beta_{i_1}} \dots S_{\beta_{i_{m-1}}}(\beta_{i_m}) < 0$. Consider

$$\begin{array}{ccccccc} \beta_{i_m}, & S_{\beta_{i_{m-1}}}(\beta_{i_m}), & S_{\beta_{i_{m-2}}} S_{\beta_{i_{m-1}}}(\beta_{i_m}), & \dots, & S_{\beta_{i_1}} \dots S_{\beta_{i_{m-1}}}(\beta_{i_m}) \\ \vee & & & & \wedge \\ 0 & & & & 0 \end{array}$$

Then $\exists k$ s.t. $S_{\beta_{i_k}} \dots S_{\beta_{i_{m-1}}}(\beta_{i_m}) = \beta_{i_{k-1}}$ and $S_{\beta_{i_{k-1}}} \dots S_{\beta_{i_{m-1}}}(\beta_{i_m}) = -\beta_{i_{k-1}} < 0$

From above, setting $u = S_{\beta_{i_k}} \dots S_{\beta_{i_{m-1}}}$, this yields $S_{\beta_{i_{k-1}}} = u S_{\beta_{i_m}} u^{-1}$

i.e. $S_{\beta_{i_{k-1}}} \dots S_{\beta_{i_{m-1}}} = S_{\beta_{i_k}} \dots S_{\beta_{i_{m-1}}} S_{\beta_{i_m}}$

$\Rightarrow w = S_{\beta_{i_1}} \dots S_{\beta_{i_m}} = S_{\beta_{i_1}} \dots S_{\beta_{i_{k-1}}} \cdot S_{\beta_{i_{k-1}}} \dots S_{\beta_{i_{m-1}}} S_{\beta_{i_m}}$ contradicting with minimality
 $= S_{\beta_{i_1}} \dots S_{\beta_{i_{k-2}}} S_{\beta_{i_k}} \dots S_{\beta_{i_{m-1}}}$

Thus $w(\beta_{i_m}) < 0$, contradicting with $w(B) = B$.

Def normal system

A set of vectors $v_1, \dots, v_k$ is called a normal system if

(1) $v_1, \dots, v_k$ are linearly independent

(2) $\|v_i\| = 1 \quad \forall i$

(3) $(v_i | v_j) \in \{0, -\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{3}}{2}\}$

Example

let $B = \{\beta_1, \dots, \beta_n\}$ be a base of a root system. Then $\{\frac{\beta_1}{\|\beta_1\|}, \dots, \frac{\beta_n}{\|\beta_n\|}\}$

Example

let $B = \{\beta_1, \dots, \beta_n\}$ be a base of a root system. Then $\left\{\frac{\beta_1}{\|\beta_1\|}, \dots, \frac{\beta_n}{\|\beta_n\|}\right\}$ is a normal system.

Def Coxeter graphs w.r.t. normal systems

vertices: v_i

edges: $4(v_i | v_j)^2$ edges between v_i and v_j

prop properties of normal systems and Coxeter graphs

(1) A subset of normal system is a normal system

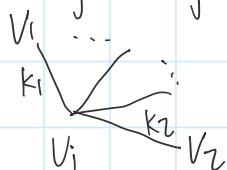
(2) let $v_1, \dots, v_k$ be a normal system. Then $\# \{ \text{connected pairs } v_i - v_j \} < k$.

pf: Set $v = v_1 + \dots + v_k \neq 0$

$$(v | v) = \sum_{i=1}^k (v_i | v_i) + 2 \sum_{i < j} (v_i | v_j) \leq k - (\# \text{ of connected pairs})$$

(3) The Coxeter graph contains no cycles (corollary of (1) & (2))

(4) Fix v_i . $\#$ of edges adjacent to v_i is < 4



pf: Note that $v_1, \dots, v_k$ must be pairwise orthogonal (otherwise we have a cycle).

Set $v = v_i - (v_i | v_1)v_1 - \dots - (v_i | v_k)v_k \neq 0$

$$\text{Then } 1 = (v_i | v_i) = (v | v) + (v_i | v_1)^2 + \dots + (v_i | v_k)^2 \Rightarrow \frac{k_1}{4} + \dots + \frac{k_k}{4} < 1$$

(5) $v_1, \dots, v_k \in$ a normal system s.t.

$$\text{i.e. } (v_i | v_j) = \begin{cases} -\frac{1}{2} & j = i+1 \\ 0 & \text{otherwise} \end{cases}$$

Then replace $v_1, \dots, v_k$ by $v_1 + \dots + v_k$, we'll get a new normal system.

pf:

1° Note that $\forall v_j \notin \{v_1, \dots, v_k\}$, v_j connects with at most one of $v_1, \dots, v_k$, otherwise there will be a cycle, hence $(v_j | v_1 + \dots + v_k) \in \{0, -\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{3}}{2}\}$

$$2^\circ (v_1 + \dots + v_k | v_1 + \dots + v_k) = k + 2 \sum_{i=1}^{k-1} (v_i | v_{i+1}) = 1$$